Requirement Analysis Phase

Introduction

The Requirement Analysis Phase is the process of identifying, understanding, documenting, and validating what users and stakeholders need from a system or project. It provides a clear foundation for designing and developing the final solution.

Objectives

- Understand user and business needs.
- Identify functional and non-functional requirements.
- Define the scope of the project clearly.
- Identify constraints, risks, and dependencies.
- Create clear requirements for the development team.

Types of Requirements

- **Functional Requirements:** Describe what the system should do, such as login, search, or report generation.
- **Non-Functional Requirements:** Describe qualities such as performance, security, reliability, and usability.
- **Business Requirements:** Define the main goals and expected outcomes of the project.
- **User Requirements:** Describe the needs and expectations of the users.

Key Steps

1. Gather requirements from users and stakeholders.
2. Analyze and organize the collected information.
3. Prioritize important requirements.
4. Document the requirements clearly.
5. Review and validate the requirements with stakeholders.

Importance

Requirement analysis reduces misunderstandings, prevents unnecessary changes, and helps the development team build a solution that meets user expectations. A clear requirement analysis also improves project planning, cost estimation, and quality.